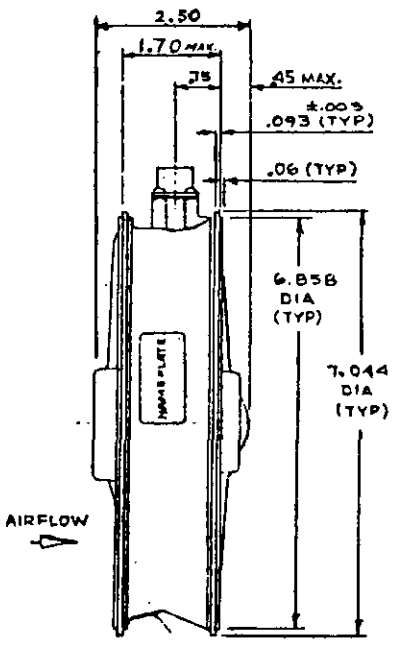
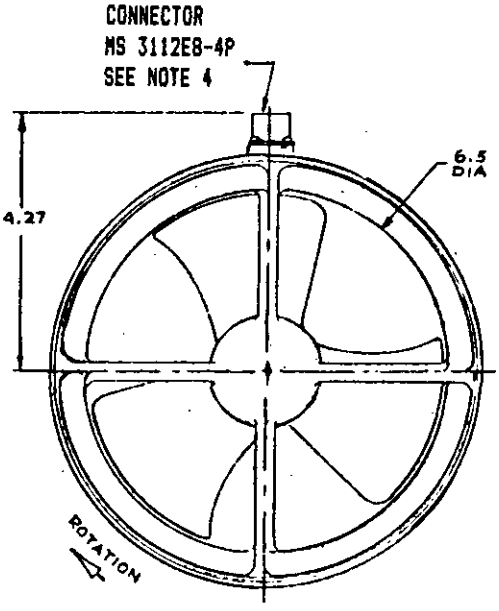
MILITARY SPECIFICATION SHEET

BLOWER, TUBEAXIAL, ELECTRONICALLY COMMUTED DIRECT CURRENT (ECDC) (BRUSHLESS) MOTOR DRIVEN - FOR COOLING ELECTRONIC EQUIPMENT (300 CFM NOMINAL)

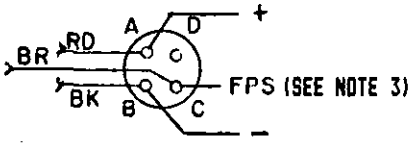
This specification is approved for use by all Departments and agencies of the Department of Defense.

The complete requirements for procuring the blower described herein shall consist of this document and the latest issue of Specification MIL-B-28873.





Internal and External Wiring



Inches	MM
.003	.08
.06	1.52
.093	2.36
.45	11.43
.75	19.05
1.70	43.18
2.5	63.50
4.27	108.46
6.50	165.10
6.858	174.19
7.044	178.92

Notes: 1. Unless otherwise specified, tolerances shall be ±.010 (.25mm) for three place decimals and ±.03 (.7mm) for two place decimals; angles ±1°.
2. Dimensions are in inches.
3. Fan Performance Sensor (FPS) shall be an optional feature of the electronics circuit.
4. Not for Air Force use; use MS 3470.

FIGURE 1. Configuration and design

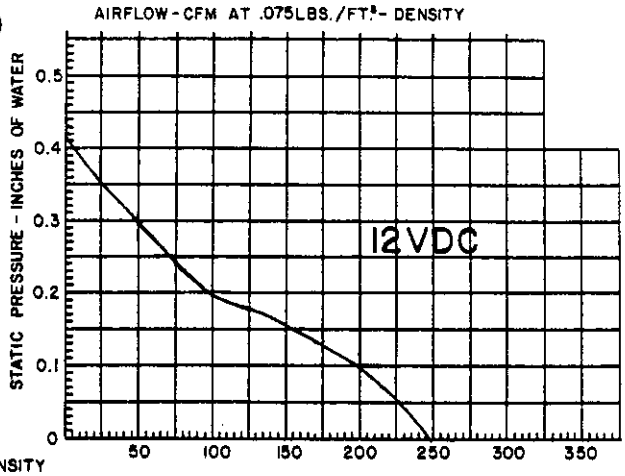
FSC 4140

DISTRIBUTION STATEMENT A.
Approved for public release; distribution is unlimited.

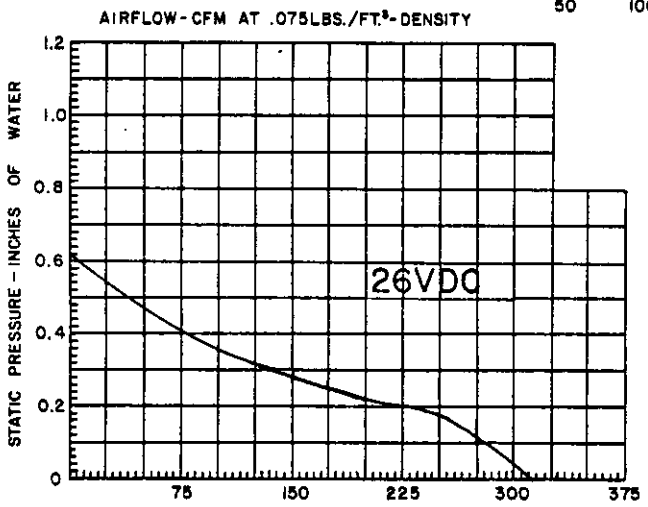
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Check the source to verify that this is the current version before use.

MIL-B-28873/3

-001



-002



-003

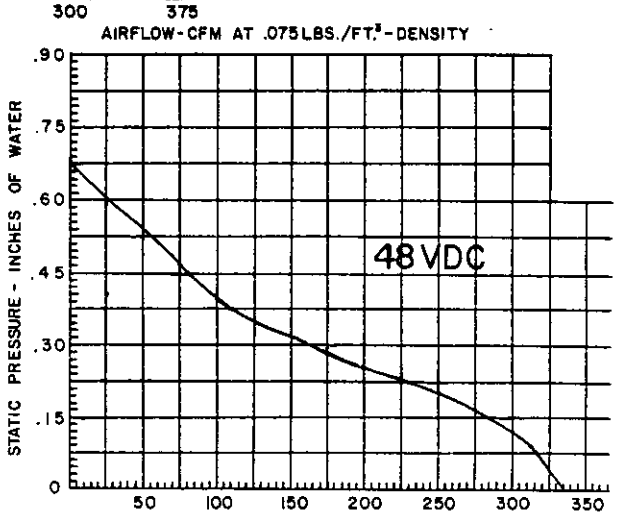


FIGURE 2. Performance curves

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Requirements:

Dimensions and configuration - See FIGURE 1.

Design and construction -

Weight: 2 pounds (.907 kilograms) maximum
Excitation: See Table I
Power consumption: See Table II
Output performance: See Table III

Cold start: -54°C ±5°C

Generated vibration: 2.8 lbs (1.27 kilograms) rms, maximum

Dynamic balance: 0.25 inches (6.35 mm) per second rms

Extended life test: See Table IV

Life expectancy: See Table V

TABLE I. Excitation

Dash	Nominal DC voltage ±10%	Duty cycle	Speed (rpm) nominal	Rated DC voltage ±2.0%
001	12	Continuous	3000	12
002	26	"	3800	26
003	48	"	4000	48

Table II Power consumption at rated excitation

Dash no.	Current run amps (max)	Power input watts (max)	Starting current amps (max)
001	.84	11.5	3.28
002	.89	25.5	3.38
003	.61	32.2	2.25

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TABLE III Output performance at rated excitation

Dash no.	Air volume @ zero SP*	Air volume @ rated SP*	Temperature rise (winding) degree C	Noise level measured @ 3 ft. (.914m) "A" weighting
	Minimum CF (m^3/s)	Min.CFM @ inch H_2O (m^3/pascal)	Max.	Decibels (dB)
001	220 (.104)	180@.1 (.085@.0025)	21	56.0
002	283 (.133)	252@.1 (.119@.0025)	45	61.0
003	303 (.143)	270@.1 (.127@.0025)	50	62.0

*Static Pressure at .075 lbs/ft^3 (1.20 kg/m^3) density

TABLE Extended life test

Dash no.	Hours	Degree C
001	1000	70
002	1000	70
003	1000	85

TABLE V. Life expentancy

Dash no.	Hours	Degree C
001	30,800	50
	28,400	65
002	25,700	50
	23,200	65
003	23,800	50
	19,700	65

Custodians:
Army - ME
Navy - EC
Air Force - 99

Review activities:
Army
Navy
Air Force - 82
DLA - GS

User Activities:
Navy - AS,SH
Air Force - 17,11

Preparing activity:
Navy - EC

Agent:
DLA - GS

(Project 4140-0128)

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